CSI132 - Discrete Structures II Tutorial 1
Attempt All questions
Duration: 2 hours
Proof Methods

1. Finish all Exercises in the Lecture slides.
2. Prove that the sum of two even integers is even.
3. Show that the additive inverse, or negative, of an even number is an even number.
4. Prove that the product of two odd numbers is odd.
5. Prove that if m and n are integers and $m.n$ is even, then m is even or n is even.
6. Prove that if n is an integer, and $3n + 2$ is even, then n is even using:
   (a) a proof by contraposition.
   (b) a proof by contradiction.
7. Prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even.
8. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.